

Arithmetic and Logic CTF

A focused reading map for the three-day exercise

Chapters 10, 11, and 12 are large, and the capture-the-flag exercise only leans on part of them. This guide points you at the sections each puzzle actually depends on, so you can spend your reading time where it pays off. It does not throw the rest away. Every chapter also has material that is not needed for these puzzles but is still part of the course, and that material is listed at the end of each day so nothing disappears on you.

How to read the priority column. **Read closely** means a puzzle cannot be solved without it. **Skim** means it gives useful background and makes the closely-read parts click, but you will not be tested by a flag on it. **Context** means read it once for the shape of the idea and move on.

1. Day 1: Number Systems and Integer Overflow

The puzzles need you to read raw bytes in binary and hexadecimal, and to understand how signed integers are stored and what happens when they overflow.

Table 1: Chapter 10 and early Chapter 11 sections for Day 1.

Section	Why the puzzles need it	Priority
Ch 10: The Binary System; Converting Between Binary and Decimal	Reading and reasoning about raw bytes	Read closely
Ch 10: Hexadecimal Notation; Table 10.3 (Decimal, Binary, Hex)	The warm-up is a hex dump; hex is how bytes are shown	Read closely
Ch 11: Integer Representation; Sign-Magnitude; Table 11.2 (4-bit representations)	Why a signed field accepts negatives, and why that is exploitable	Read closely
Ch 11: Twos Complement addition (Fig 11.3) and the Overflow Rule	The 16-bit overflow puzzle is this rule in action	Read closely
Ch 11: Negation and the two Negation Special Cases	The most-negative-value quirk, used as an extension	Skim
Ch 11: Range Extension; Fixed-Point Representation	Rounds out integer storage	Context
Ch 10: Decimal and binary fractions (Figs 10.1, 10.2)	Method for fractions; not used until Day 3 in spirit	Context

On the syllabus, not needed for Day 1 puzzles: positional number systems in other bases such as base 7 (Table 10.2), and the unsigned and twos complement multiplication and division methods, including Booth's algorithm and restoring division (Figures 11.7 through 11.17). Read

these for the course; no flag depends on them.

2. Day 2: Logic, Gates, and XOR

The puzzles need the XOR truth table and its reversibility, the basic gates and Boolean identities, and enough of sum-of-products and Karnaugh maps to simplify a small function you recover from compiled code.

Table 2: Chapter 12 sections for Day 2.

Section	Why the puzzles need it	Priority
Boolean Variables and Operations; Table 12.1 (Boolean Operators)	The XOR operation, its truth table, and the identity $a \oplus a = 0$	Read closely
Fig 12.1 (Basic Boolean Functions of Two Variables)	Seeing XOR alongside AND and OR as a gate	Read closely
Fig 12.3 (Basic Logic Gates); Table 12.3 (Basic Identities)	Recognizing AND, OR, NOT in decompiled code and simplifying them	Read closely
Combinational Circuit; Table 12.4; Fig 12.6 (Sum of Products)	Writing a function as a sum of product terms, which is what the logic lock is	Read closely
Figs 12.9 through 12.11 (Karnaugh Maps)	An optional way to check the recovered function is fully simplified; not needed to solve anything	Skim
Fig 12.7 (Product of Sums); Fig 12.8 (Simplified Implementation)	The dual view and a worked simplification	Skim
Table 12.2 (Boolean algebra and set operations); Fig 12.2 (Venn diagram)	Intuition for the operators	Context

A word on Karnaugh maps. Karnaugh maps take practice, and they can be confusing the first few times you use one. Do not worry if they do not make sense right away. You will not need one to finish any of the challenges, so treat a Karnaugh map as something to explore, not something to master before we start.

On the syllabus, not needed for Day 2 puzzles: the Quine-McCluskey method (Tables 12.6 and 12.7), and the standard combinational building blocks that come after the gate material, namely multiplexers, decoders, read-only memory, and the binary adder (roughly Figures 12.14 through 12.23). Worth knowing for the course; no flag depends on them.

3. Day 3: Floating Point and Sequential Logic

The puzzles need the idea that floating point is an approximation with special values, especially NaN and its comparison behavior, and the shift register plus flip-flop material that a linear feedback shift register is built from.

Table 3: Chapter 11 floating point and Chapter 12 sequential sections for Day 3.

Section	Why the puzzles need it	Priority
Ch 11: Floating-Point Representation; Fig 11.18 (32-bit format)	What the bits mean and why values are approximate	Read closely
Ch 11: Table 11.7 (Operations that Produce a Quiet NaN) and NaN behavior	The safe puzzle turns on NaN comparisons all being false	Read closely
Ch 11: Precision Considerations	The Patriot missile story is this section made concrete	Skim
Ch 12: Sequential Circuit; Flip-Flops; Fig 12.27 (D Flip-Flop)	A shift register is a chain of flip-flops	Read closely
Ch 12: Fig 12.31 (5-Bit Shift Register)	The LFSR is exactly this plus one XOR tap	Read closely
Ch 11: IEEE 754 formats and Table 11.5 (interpretation)	The full standard around the special values	Skim
Ch 12: Fig 12.24 (S-R Latch); Fig 12.28 (J-K Flip-Flop)	Other flip-flop types for background	Context

On the syllabus, not needed for Day 3 puzzles: floating-point addition, subtraction, multiplication, and division procedures and guard bits (Figures 11.22 through 11.25), subnormal numbers (Figure 11.26), and the remaining sequential and programmable-logic topics, namely counters, the programmable logic array, and the FPGA (roughly Figures 12.33 through 12.36). Course material; no flag depends on them.

4. The one-paragraph version

If you read nothing else: for Day 1, binary and hexadecimal from Chapter 10 plus signed representation and the overflow rule from Chapter 11. For Day 2, the XOR operation, the basic gates and identities, sum of products, and Karnaugh maps from Chapter 12. For Day 3, the floating-point representation and NaN behavior from Chapter 11 plus flip-flops and the shift register from Chapter 12. Everything else in the three chapters is still yours to learn for the course, and it is listed above so you know exactly what you are setting aside for now.